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Odds ratio based complementary-logic particle swarm optimization for multiple gene-gene interactions in breast cancer susceptibility

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Abstract

The single nucleotide polymorphism (SNP) pattern has a significant impact on an individual's susceptibility to disease. However, it is currently difficult to fully analyze the association of genotype frequencies of cases and controls because the interaction of SNPs has not been addressed in recent studies. We applied an improved particle swarm optimization algorithm to analyze the association of genotype frequencies between case and control data. The high-dimensional simulated genotype data from real SNP genotype frequencies was used to test the capability of the proposed method. We systematically evaluated the joint effect of 27 SNP combinations of ten published epigenetic modifier-related genes involved in breast cancer related pathways. The results show that an optimal gene-gene interaction combination with a maximal difference between controls and cases can be predicted with the improved PSO method. Finally, we used the odds ratio to effectively analyze the potential breast cancer risk of the predicted gene-gene interaction combinations.

1 Introduction

Single nucleotide polymorphisms (SNPs) constitute an abundant form of genetic variations amongst species. The association relationship amongst SNPs, diseases and cancers, as well as their therapies and pharmacogenomics, have been studied and reviewed in previously published literature reports [1-5]. Currently, many researchers focus their attention on gene variations associated with hereditary phenotypes. The key issue in this field is to identify the significant and representative gene-gene associations of the large number of candidate genes and SNPs that are related to diseases or cancers.

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Many approaches propose computational methods to determine the association of a disease with multiple SNPs simultaneously; these methods include modeling interactions statistically [6], combinatorial partitioning, neural networks (NN) [7], and multifactor dimensionality reduction (MDR) [8]. The MDR method, which uses binary outcomes to classify results into high-risk and low-risk groups, has a strong prediction capability and can detect and characterize the high-order gene–gene interactions in case–control studies. However, the MDR method does not allow a comparison of the disease risks between different combinations when genotypes are analyzed.

A new and improved method for exploring the multiple gene interaction problems is thus required. An evolutional algorithm can be employed to identify an optimal gene-gene combination from the vast amount of possible combinations. We hypothesize that interactions between polymorphisms of these genes may have a synergistic or non-additive effect on the pathogenesis of a disease, and therefore may be used to explain differences in the disease risk. Each genotype has three possible SNP combinations: AA, Aa, and aa for a SNP with a major/minor polymorphism. Accordingly, each individual has its own SNP pattern for each SNP. We used an improved particle swarm optimization (PSO) to solve the optimization problems associated with gene-gene interactions. PSO has properties of randomized search and is an optimization technique that derives its working principles from simulations of the social behavior of organisms. Results demonstrate that this method can find the best gene-gene interaction with maximal difference between controls and cases. The odds ratio of each combination of genotypes was regarded as a quantitative measure of the disease risk.

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2 Methods

We propose the OR-CPSO (odds ratio based complementary-logic particle swarm optimization) method to generate the best gene-gene interaction with genotypes to predict disease susceptibility. OR-CPSO is implemented in two stages. Stage 1 consists of the CPSO method itself, and in stage 2, the odds ratio is used to determine the best combination of genotypes and quantitatively measure the disease risk.

Particle Swarm Optimization (PSO)

Particle swarm optimization (PSO) was developed by Kennedy and Eberhart in 1995 [9]. PSO is based on a population stochastic optimization technique which simulates the social behavior of organisms, such as birds in a flock or fish in a school. Each individual bird or particle can make use of its own memory and knowledge gained by the entire swarm to find an optimal position. All particles own two important properties: (i) a fitness value, which is determined by a designed fitness function, and (ii) a velocity, which affects the movement of the particle. During the search, each particle adjusts its position according to its own experience and the experience of a neighboring particle. Finally, the particles follow the current best particle in the search space until a predefined number of iterations is reached. The flowchart of the method is shown in Fig. 1.

In PSO, the position of the *i*th particle can be represented as $x_i = (x_{i1}, x_{i2}, \ldots, x_{iD})$, in which D represents the dimension of the search space. The velocity of the *i*th particle can be represented as $v_i = (v_{i1}, v_{i2}, \ldots, v_{iD})$. The positions X and velocities V of the particles are confined within $[X_{\min}, X_{\max}]^D$ and $[V_{\min}, V_{\max}]^D$, respectively. The best currently visited position of the *i*th particle is denoted as its individual best position $p_i = (p_{i1}, p_{i2}, \ldots, p_{iD})$, this vector is called $pbest_i$. The $pbest_i$ with the best fitness amongst all individuals is denoted the global best position $g = (g_1, g_2, \ldots, g_D)$ and called *gbest*. The position and velocity of the *i*th particle are updated via $pbest_i$ and *gbest* in the swarm. The parameters w, r_1 and r_2 are key factors affecting the convergence behavior [10, 11]. The inertia weight w controls the balance between the global exploration and the local search ability. A large w favors global search, whereas a small w favors local search. A w that linearly decreases from 0.9 to 0.4 throughout the search process is widely used [12].

Complementary-logic Particle Swarm Optimization (CPSO)

The complementary-logic strategy aims to assist particles trapped in a local optimum by moving them to a new region in the search space. This ensures a global exploration of the search space. In standard PSO, particles may get trapped in a local optimum due to the premature convergence of particles. Each particle's position is randomly initialized in the search space and updated by the following two values: *pbest*, which is the best fitness the particle itself has achieved so far, and *gbest*, which is the best fitness of all particles in the population so far. However, based on this update equation, each particle only moves in the vicinity of *gbest* and the distance between *gbest* and the neighboring particles is thus very small. For this reason, complementary particles are introduced in order to avoid premature convergence. These new particles replace a randomly selected 50% of the particles in the swarm [15]. The complementary particles are essential for the success of a given optimization task. In this study, a complementary-logic adaptive population of PSO is used to search for gene-gene interaction combinations. The entire process consists of (1) initialization of the particle swarm, (2) fitness evaluation, (3) identification of *pbest* and *gbest*, (4) termination criteria, (5) particle updating, and (6) complementary-logic adaption of particles.

(1) Initialization of particle swarm

In CPSO, each particle in a population is associated with a solution group. The representation of a particle can be divided into two parts: the number of selected SNPs, and the genotype associated with the SNPs. A particle can thus be represented as follow:

$$C_i = (S_{ij}, G_{ij}),\ i=1,2,...,m,\ j=1,2,...,n \qquad (1)$$

where m represents the size of the population and n represents the number of SNPs selected. A gene thus described represents the selection of the *j*th SNP on the *i*th particle; S_{ij} represents the selected SNP; G_{ij} represents the three possible genotypes (AA, Aa, aa) once S_{ij} is selected. Since an SNP can not be repeatedly selected, the S_{ij} selection can not be repeated, and thus no restrictions are imposed on genotype (G_{ij}) selection.

The size of each particle can be changed based on the number of SNPs selected. For example, a particle with a total of 8 SNPs can be represented by randomly choosing four SNPs and four different genotypes, for instance, $C_i = \{1,3,5,7,1,2,2,1\}$. In this representation of the particle the first four numbers represent the chosen SNPs {1,3,5,7}, and the second four numbers represent the chosen genotypes {1,2,2,1} in which the 1,2, and 3 represent genotype AA, Aa, and aa. The selected SNPs and SNPs associated with the genotypes were randomly generated. In the above case, they were the following: (1, 1), (3, 2), (5, 2), and (7, 1).

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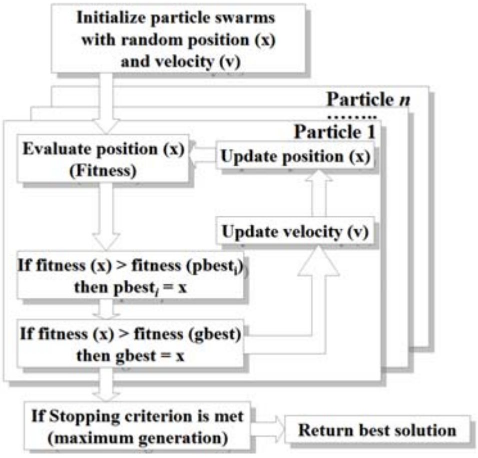

Fig. 1. Particle swarm optimization flowchart.

(2) Fitness evaluation

The fitness values of SNP combinations (gene-gene interaction with genotype) have to be computed for all selected SNPs, whereas chromosomes of a different size can not be included in the computation of the evolution process. Thus, the number of SNPs has to be set to focus on SNP combinations that obtain the highest fitness value. To determine the maximum fitness value, we divided the fitness calculation into two steps. Firstly, the amount of combined SNPs with genotypes in the control dataset is calculated. Secondly, the amount of combined SNPs with genotypes in the case dataset is calculated. Eq. (2) is the used to determine the fitness value of each chromosome. A high fitness value represents a high breast cancer risk for an SNP selection and genotype combination. Each particle is calculated using the fitness function shown below:

$$f(Ci) = \frac{|Control\_match - Case\_match|}{ALL\_Control + ALL\_Case} \times 100\% \qquad (2)$$

where *ALL_Control* is the number of members in the control data; *ALL_Case* is the number of members in the case data; *Control_match* is the number of members that match the selection conditions in the control data; *Case_match* is the number of members that match the selection conditions in the case data. The difference of the number of members in control and case can be used to calculate the percentage of members in each data category. A high percentage means that a haplotype of the SNP selection and genotype combination is more likely associated with the occurrence of breast cancer.

(3) Identification of *pbest* and *gbest*

One of the characteristics of PSO is that each particle has a memory of its own best experience. Each particle finds its personal best position and velocity (*pbest*) and the global best position and velocity (*gbest*) when moving. If the fitness value of a particle P_v in the current iteration is better than the fitness value of *pbest* in the previous iteration, *pbest* is updated to P_v. If the fitness value of a particle P_v in the current iteration is better than *gbest* in the previous iteration and is the best one in the current iteration, *gbest* is updated to P_v. Each particle then adjusts its direction based on *pbest* and *gbest* in the following iteration.

(4) Judgment of termination condition

The method is terminated when *gbest* has achieved the best position or when a maximum number of iterations is reached. When the termination condition is reached, *gbest* constitutes the optimal solution. The maximum number of iterations is described in the parameter settings section.

(5) Particle update

Each particle adjusts its direction in the following iteration via an evaluation of *pbest* and *gbest*. The update equations are formulated in Eqs. (3) to (5):

$$w_{LDW} = (w_{\max} - w_{\min}) \times \frac{Iteration_{\max} - Iteration_i}{Iteration_{\max}} + w_{\min} \qquad (3)$$

$$v_{id}^{new} = w_{\mathrm{LDW}} \times v_{id}^{old} + c_1 \times r_1 \times \left(pbest_{id} - x_{id}^{old}\right) + c_2 \times r_2 \times \left(gbest_d - x_{id}^{old}\right) \qquad (4)$$

$$x_{id}^{new} = x_{id}^{old} + v_{id}^{new} \qquad (5)$$

In Eq. (3), $w_{\max}$ is 0.9, $w_{\min}$ is 0.4 and $Iteration_{\max}$ is the maximum number of allowed iterations [13]. In Eq. (4), r_1 and r_2 are random independent numbers between (0, 1); c_1 and c_2 are acceleration coefficients equal to 2 [13]; these coefficients constantly controls how far a particle moves in a single iteration.

(6) Complementary-logic adaption in particles

The principal task of the complementary-logic is to free particles trapped in a local optimum and moving them to a new region of the search space, thus ensuring global exploration. The position of a trapped particle is changed based on the following complementary rules:

$$x_{id}^{Complement} = (x_{\max} + x_{\min}) - x_{id}^{selected} \qquad (6)$$

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CPSO pseudo-code

```
begin
  Randomly initialize particle swarm
  while (the stopping criterion is not met)
    Adjust position of particle swarm
    Evaluate fitness of particle swarm
    for i = 1 to number of particles
       Find pbest_i
       Find gbest
       for d = 1 to number of dimension with particle
         update the position of particles by Eqs.4-5
       next d
    next i
    Update the inertia weight value by Eq.3
    if fitness of gbest is the same ten times then
      Randomly select 50% of the particles of swarm S
      Generate new particles C via Eq. 6 and replace S
    end if
  next generation until stopping criterion
end
```

3 Results

Dataset

The dataset was obtained from the epigenetic modifiers pathway (76 SNPs for 10 genes) in the breast cancer association study. Pharoah et al. provided the genotype frequencies rather than the original raw data for the genotypes [14]. However, the original data involves different numbers of genotypes, and hence we had to normalize each genotype size to allow further analysis. Our simulated data was randomly generated and thus obey the original genotype frequency in the whole dataset. We assume that SNP_a is rs535586 in the original data, and SNP_b is the simulated data after normalization. Suppose the amount of three genotypes in SNP_a is 4547, in which AA, Aa and aa are 2139, 1908 and 500, respectively. First, we calculated the percentage of each genotype in SNP_a, i.e., 2139/4547 (47%) for AA, 1908/4547 (42%) for Aa and 500/4547 (11%) for aa. Then, based on this percentage, the modified data for SNP_b is obtained by multiplication of the percentage with the amount of the complete dataset, i.e., 47% × 5000=2353 for AA, 42%×5000=2098 for Aa and 11%×5000=549 for aa. The simulated data for SNP_b has thus been normalized to 5000 (2353+2098+549=5000). All original data were normalized to the same number in this manner. The normalization procedure is shown in the "Pseudo-code for randomly generated data" below.

Pseudo-code for randomly generated data

```
begin
  Set size = 5000
  Set number of genotype = 3
  Calculate amount of three genotypes
  while (all SNPs are not normalized)
    Calculate amount of each genotype
    Calculate numbers of each normalized genotype
    for n = 1 to number of genotype
       Randomly create numbers of each normalized
       genotype
    next n
end
```

Identification of the best gene-gene interaction with a maximum difference between cases and controls

We computed all differences between cases and controls among the combinations. For example, the SNP1-genotype "Aa" and SNP11-genotype "Aa", showed the maximal difference between case and control groups amongst the 2-SNP combinations. In the same manner the best performance of 3-5 SNP-combinations are calculated.

The relative strength of the influence of an SNP on breast cancer was determined by CPSO. The results for 2-5 SNP combinations are shown in Table 2. In the 2-SNP combination, the interaction is SNP (1, 11) with genotype "Aa-Aa". In the 3-SNP combination, the interaction is SNP (1, 9, 11) with genotype "Aa-AA-Aa". In the 4-SNP combination, the interaction is SNP (1, 8, 9, 11) with genotype "Aa-AA-AA-Aa". In the 5-SNP combination, the interaction is SNP (1, 2, 8, 9, 11) with genotype "Aa-AA-AA-AA-Aa". These are the same combinations as in the preceding section and thus CPSO provided the highest difference in terms of gene-gene interaction between the control and case groups for a fixed number of SNPs.

Ranks of the odds ratios indicate the highest risk for breast cancer

The breast cancer susceptibility of multiple-combination epigenetic modifier-related gene SNPs was analyzed. Odds ratios (OR) are widely used in medical reports and offer a special and very convenient interpretation in case-control studies when the risk ratio can not be obtained directly. The OR is an alternative to the correlation coefficient as a measure of association between case and control samples. It has many desirable properties and is possibly easier to interpret than the correlation coefficient. Table 2 shows the estimated effect (odds ratio, 95% CI) of certain SNP barcodes with respect to breast cancer susceptibility. Patients with specific SNP combinations (two, three, four, and five SNPs), i.e., SNPs (1, 11), SNPs (1, 9,11), SNPs (1, 8, 9, 11), and SNPs (1, 2, 8, 9, 11), had 1.159 (CI: 1.049-1.281, p= 0.004), 1.175 (CI: 0.035-1.334, p= 0.014), 1.194 (CI: 1.010-1.411, p= 0.037) and 1.186 (CI: 0.955-1.473, p= 0.136) have reduced odds of developing breast cancer, respectively, when compared to other SNP-genotype combinations.

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4 Discussion

Contribution of this work to gene-gene interaction research

An SNP from a single susceptible gene may have only a marginal effect and may not be detected by traditional statistical analyses since typically multiple genes and their SNPs are involved in or associated with complex diseases and cancers. These studies focus on the interaction of chromosome-wide SNPs, which are located in different chromosomes. Although the applicable studies may find that each SNP has only a moderate association, they do not analyze multiple SNP combinations because the association may be stronger. Since multiple SNP combinations can have a higher impact on the development of breast cancer, association studies involving multiple SNPs from genes related to multiple diseases have gained in popularity. The objective is to understand what contributions to the breast cancer risk come from functionally relevant joint effects of combinational SNPs within and between different cancer pathways. SNPs of 27 related genes from different chromosomes were selected to simultaneously investigate gene-gene interaction related to breast cancer. These combined SNP-genotypes show that better protection against breast cancer can suggest the cumulative association of genetic variants to cancers. However, susceptibility to cancer or disease in relation to many previously unidentified gene-gene interactions should be considered simultaneously.

Analysis of gene-gene interaction

The results list the relative associations of combinations of 2-5 SNPs with genotypes in order to understand the occurring of breast cancer, i.e., the maximal occurrence difference between controls and cases. The OR-CPSO method utilizes ternary genotypes rather than the commonly used binary allele types. This improves the resolution of the multiple gene–gene interaction analysis. The odds ratio is an important tool used to interpret the results since it only focuses on the evaluation of gene–gene associations to a breast cancer predisposition rather than prove the direct evidence to breast cancer risk. Carriers of the best SNP combination were compared with all other individuals via logistic regression, and hence combination must show the lowest odds ratio in our case. In addition, other combinations inversely associated with a breast cancer risk are also included.

Advantages of the complementary-logic PSO algorithm

In general PSO, each individual, together with the population as a whole, evolves towards a best fitness value, which is determined by an objective function. Although this increases the convergence capability, convergence often occurs too fast, and results in the population getting stuck in a local optimum as the swarm's diversity rapidly decreases. Complementary-logic is a very powerful technique for avoiding entrapment in local optima since the diversity of the particle swarm is maintained and the complexity is not increased. The time complexity $\Theta(D)$ of CPSO is the same as for PSO. D denotes the number of the initial feature sets, and $\Theta(\ )$ denotes a tight estimate of the complexity.

Fig. 2 illustrates the advantage of the complementary-logic technique. In Fig. 2-A, we assume that *gbest* has not been updated for ten iterations. Under these circumstances 50% [15] of the particles are selected to become complementary particles C, i.e., S_1, S_2 and S_3. The complementary position of these particles is calculated with Eq. 6. Fig. 2-B shows how the new positions of the selected S particles correspond to the complementary particles. Fig. 2-C describes how the complementary particles continue their search in new regions of the search space. The most ideal situation is shown in Fig 2-D, in which particles find better solutions and the search direction of the swarm is changed towards a more optimal region of the search space.

5 Conclusion

Previous literature reports have focused only on design comparisons for case-control studies with a single SNP. In this study, multiple SNPs are exclusively used in association studies to investigate polygenic diseases and cancers. In nature, genes can be damaged or modified in various ways. This damage can lead to an increased risk of developing many diseases or cancers. It is thus important to obtain informative SNP patterns of the SNPs located in the involved genes and the related pathways. Due to the high number of SNPs involved, analysis of association studies is difficult to perform, especially when multiple SNPs are investigated simultaneously. We propose an improved PSO to perform a powerful analysis of 27 SNP cross-interactions, and provide representative gene-gene interactions for breast cancer. The risk of developing breast cancer due to the gene-gene interactions is evaluated by the odds ratio. The results demonstrate that CPSO is capable of identifying the best fitness of cases and controls and can potentially be applied to determine the complex gene-gene SNP interactions amongst the huge number of SNPs involved in genome-wide association studies.

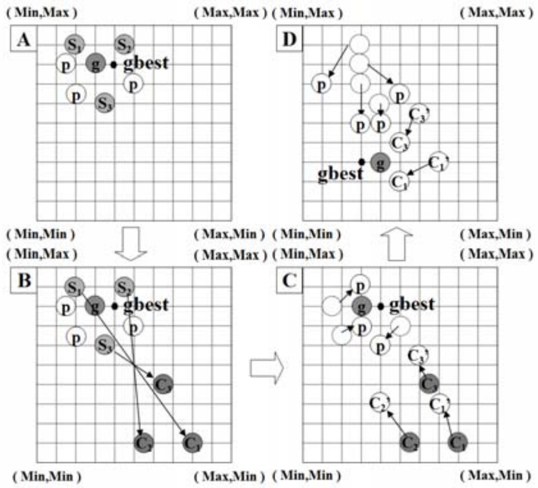

Fig. 2. Creation of complementary particles. Particles S represent the selected particles, which are subsequently replaced by their respective complementary particles C.

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Table 1. SNPs investigated in this study and their normalized numbers of polymorphism genotyping information.

SNP No.	Gene and SNP	Genotype AA (Control /Cases)	 Aa (Control /Cases)	 aa (Control /Cases)	SNP No.	Gene and SNP	Genotype AA (Control /Cases)	 Aa (Control /Cases)	 aa (Control /Cases)
1	BAT8 (rs535586)	2353 2208	2098 2213	549 579	15	DNMT3B (rs2424932)	1788 1849	2406 2363	806 788
2	BAT8 (rs652888)	3079 2995	1676 1748	245 257	16	DNMT3B (rs2424928)	1546 1515	2440 2447	1014 1038
3	DNMT1 (rs2290684)	1251 1280	2534 2656	1215 1155	17	DNMT3B (rs992472)	1871 1878	2351 2354	778 768
4	DNMT3A (rs7587636)	1412 1425	2486 2506	1102 1069	18	DNMT3B (rs2424913)	1539 1536	2460 2469	1001 995
5	DNMT3A (rs6749992)	1395 1355	2442 2477	1163 1168	19	DNMT3B (rs1333469)	1733 1799	2405 2340	862 861
6	DNMT3A (rs749131)	1568 1579	2466 2481	966 940	20	EHMT1 (rs4526432)	1544 1560	2460 2436	996 1004
7	DNMT3A (rs7560488)	1747 1744	2398 2406	855 850	21	EHMT1 (rs6559218)	2035 2056	2385 2267	580 677
8	DNMT3A (rs734693)	2726 2765	1925 1885	349 350	22	EHMT1 (rs7852475)	1968 2003	2340 2314	692 683
9	DNMT3A (rs2276599)	2745 2784	1901 1874	354 342	23	HDAC2 (rs352063)	2051 2067	2293 2324	656 609
10	DNMT3A (rs2289195)	1628 1706	2477 2425	895 869	24	HDAC2 (rs3778216)	2695 2664	1971 1974	334 362
11	DNMT3A (rs7581217)	2152 2010	2213 2362	635 628	25	MBD2 (rs1259936)	2147 2146	2239 2267	614 587
12	DNMT3A (rs2304429)	1703 1698	2428 2387	869 915	26	MTHFR (rs1801133)	2237 2286	2217 2105	546 609
13	DNMT3A (rs6722613)	1656 1644	2429 2513	915 843	27	SETDB1 (rs4970986)	2111 2163	2313 2250	576 587
14	DNMT3B (rs6058897)	1262 1322	2480 2494	1258 1184					

Table 2. CPSO-generated best SNP combinations and the estimated effect of the gene-gene interactions on the occurrence of breast cancer.

Combined SNPs	SNP genotypes	Cases	Controls	Difference	Odds ratio	95% CI
2-SNPs SNP(1,11)	**Aa-Aa** other	**1010** 3990	**896** 4104	114	1.159	1.049-1.281 (0.004)
3-SNPs SNP(1,9,11)	**Aa-AA-Aa** other	**575** 4425	**498** 4502	77	1.175	1.035-1.334 (0.014)
4-SNPs SNP(1,8, 9,11)	**Aa-AA-AA-Aa** other	**319** 4681	**270** 4730	49	1.194	1.010-1.411 (0.037)
5-SNPs SNP(1,2,8,9,11)	**Aa-AA-AA-AA-Aa** other	**184** 4816	**156** 4844	28	1.186	0.955-1.473 (0.136)